



SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT

COURSE INFORMATION SHEET
(For Theory Based Course)

Session:	Spring-2024
Course Title:	Introduction to Operating System
Course Code:	CS-222T
Credit Hours:	3+0
Semester:	4 th (A, B, C, D, E)
Pre-Requisite:	Data Structures & Algorithms (CS212T)
Instructor Name:	Mr. Adeel Saeed , Mr. Nawaid Hasan
Email and Contact Information:	adsaeed@ssuet.edu.pk, HS-12 (Cubical 2)
WhatsApp Group	OS_spring2024
Office Hours:	Wednesday & Thursday (10:00am to 11:00am)
Mode of Teaching:	Synchronous/Asynchronous/ Hybrid/Blended

COURSE OBJECTIVES:

This course covers the concepts related to operating system, its services and the design of an operating system, the structure and organization of the file system, what a process is and how processes are synchronized and scheduled, different approaches to memory management.

COURSE OUTLINE:

Operating systems basics, system calls, process concept and scheduling, inter-process communication, multithreaded programming, multithreading models, threading issues, process scheduling algorithms, thread scheduling, multiple-processor scheduling, synchronization, critical section, semaphore, synchronization problems, deadlocks, detecting and recovering from deadlocks, memory management, swapping, contiguous memory allocation, segmentation & paging, virtual memory management, demand paging, thrashing, virtual machines, Distributed operating systems , network and distributed system structures , distributed file system.

COURSE LEARNING OUTCOMES (CLOs) and its mapping with Program Learning Outcomes (PLOs):

CLO No.	Course Learning Outcomes (CLOs)	PLOs	Bloom's Taxonomy
1	Explain fundamental operating system concepts such as processes, threads, CPU scheduling, deadlock.	PLO_2 (Knowledge for solving computing problem)	C2 (Understand)
2	Apply different CPU scheduling algorithms and deadlock avoidance algorithm.	PLO_3 (Problem Analysis)	C3 (Apply)
3	Demonstrate process schronization, memory management including paging, segmentation and virtual memory.	PLO_4 (Design of solution)	C3 (Apply)



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RELATIONSHIP BETWEEN ASSESSMENT TOOLS AND CLOs:

Assessment Tools	CLO-1	CLO-2	CLO-3
Quizzes	3(10%)	3(10%)	4(11%)
Assignments	3(10%)	3(10%)	4(11%)
Midterm Exam	15(48%)	15(48%)	--
Final Exam	10(32%)	10(32%)	30(79%)
Total	31(31%)	31(31%)	38(38%)

GRADING POLICY:

Assessment Tools	Percentage
Quizzes	10%
Assignments	10%
Midterm Exam	30%
Final Exam	50%
TOTAL	100%

Relevant Jobs /Skills:

- Network Engineer
- Cloud Solutions Architect
- DevOps Engineer

Recommended Books:

- Silberschatz, Abraham. Operating System Concepts, 9th Edition. Austin, John Wiley and Sons, 2013.
<http://www.cs.nthu.edu.tw/~ychung/slides/CSC3150/Abraham-SilberschatzOperating-System-Concepts---9th2012.12.pdf>

Reference Books:

- Stallings, William. Operating Systems: Internals and Design Principles, 7th Edition. US, Prentice Hall, 2009.
https://repository.dinus.ac.id/docs/ajar/Operating_System.pdf
- Tanenbaum, Andrew. Modern Operating Systems, 3rd Edition. UK, Prentice Hall, 2008.
<http://stst.elia.pub.ro/news/SO/Modern%20Operating%20System%20%20Tanenbaum.pdf>

Benchmark Universities:

International University: <https://www.tru.ca/distance/courses/comp3411.html>

National University: <https://libguides.riphah.edu.pk/c.php?g=400821&p=2729891>



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LECTURE PLAN

Course Title: Introduction to Operating System

Course Code: CS-222T

Week No.	Week Dates	Topics	Required Reading	Key Date
1	12-03-24 15-03-24	Computer System Structure Computer System Organization OS Basics Types of OS (Single User, Batch, Multiprogramming, Time Sharing, Real Timesharing) Interrupts, Traps.	Silberschatz CHAP-1 pg. 4 – 33	
2	18-03-24 22-03-24	OS components OS services OS kernel System calls	Silberschatz CHAP-2 pg. 55 – 78	Assignment # 1
3	25-03-24 29-03-24	Process (States, PCB, Types of Schedulers) Process creation and termination Commands (fork) Cooperating processes Producer consumer problem Inter Process Communication and process synchronization	Silberschatz CHAP-3 pg. 105 – 133	
4	01-04-24 05-04-24	Threads Single and multithreaded process Thread Scheduling Models of threads P thread library (create thread) Thread example code	Silberschatz CHAP-4 pg. 163 – 189	Quiz # 1
5	08-04-19 12-04-24	CPU Scheduling Short term scheduler Dispatcher Scheduling algorithms (FCFS,SJF) Scheduling algorithms (SRTF, RR)	Silberschatz - CHAP-6 pg. 261 - 285	
6	15-04-19 19-04-24	Scheduling algorithms multilevel queues Multilevel feedback scheduling Estimation of next CPU burst	Silberschatz - CHAP-6 pg. 300 - 311	Assignment # 2
7	22-04-24 26-04-24	Deadlock System Model Deadlock Prevention Deadlock Avoidance	Silberschatz - CHAP-7 pg. 315 - 337	Quiz # 2



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8	29-04-24 03-05-24	Deadlock System Model Deadlock Prevention	Silberschatz - CHAP-7 pg. 315 - 337	
Midterm Examination (06-05-24 to 11-05-24)				
10	13-05-24 17-05-24	Process Synchronization Conditions for the good solution of critical section	Silberschatz - CHAP-5 pg. 219 - 237	
11	20-05-24 24-05-24	2-process critical section (Peter's solutions)	Silberschatz - CHAP-5 pg. 219 - 237	
12	27-05-24 31-05-24	Process Synchronization n-process critical section (Bakery Algorithm)	Silberschatz CHAP-5 pg.238 – 242	Assignment # 3
13	03-06-24 07-06-24	Main Memory Best Fit ,First Fit, Worst Fit Algorithms	Silberschatz CHAP-8 pg. 360 – 364	
14	10-06-24 14-06-24	Main Memory(cont.) Contiguous Memory Allocation Paging	Silberschatz CHAP-8 pg. 360 – 364	
15	17-06-24 21-06-24	Paging(cont.) Structure of the Page Table Segmentation Structure of the Page Table	Silberschatz CHAP-8 pg.366– 378	Quiz # 3
16	24-06-24 28-06-24	Demand Paging Virtual Memory Virtual Machines (Design, issues and implementations)	Silberschatz - CHAP-9 pg. 397 - 422	
17	01-07-24 05-07-24	Distributed operating systems Network and distributed system structures	Silberschatz - CHAP-17 pg 741- 770	
18	08-07-24 12-07-24	Distributed coordination Remote services Distributed File Systems	Silberschatz - CHAP-17 pg 741- 770	
Final Examination (15-07-24 to 26-07-24)				

Teacher's Name and Signature: Adeel Saeed

Date: 28-03-2024

Chairperson's Name and Signature: Dr. Waleej Haider

Date: 28-03-2024